

SerialFlow Studio User Guide

Professional USB-to-serial communication and logging tool | Version v1.0.1 | May 19, 2026

Purpose: This guide helps a user install, launch, connect, monitor serial data, send commands, record logs, and build the Windows executable for SerialFlow Studio.

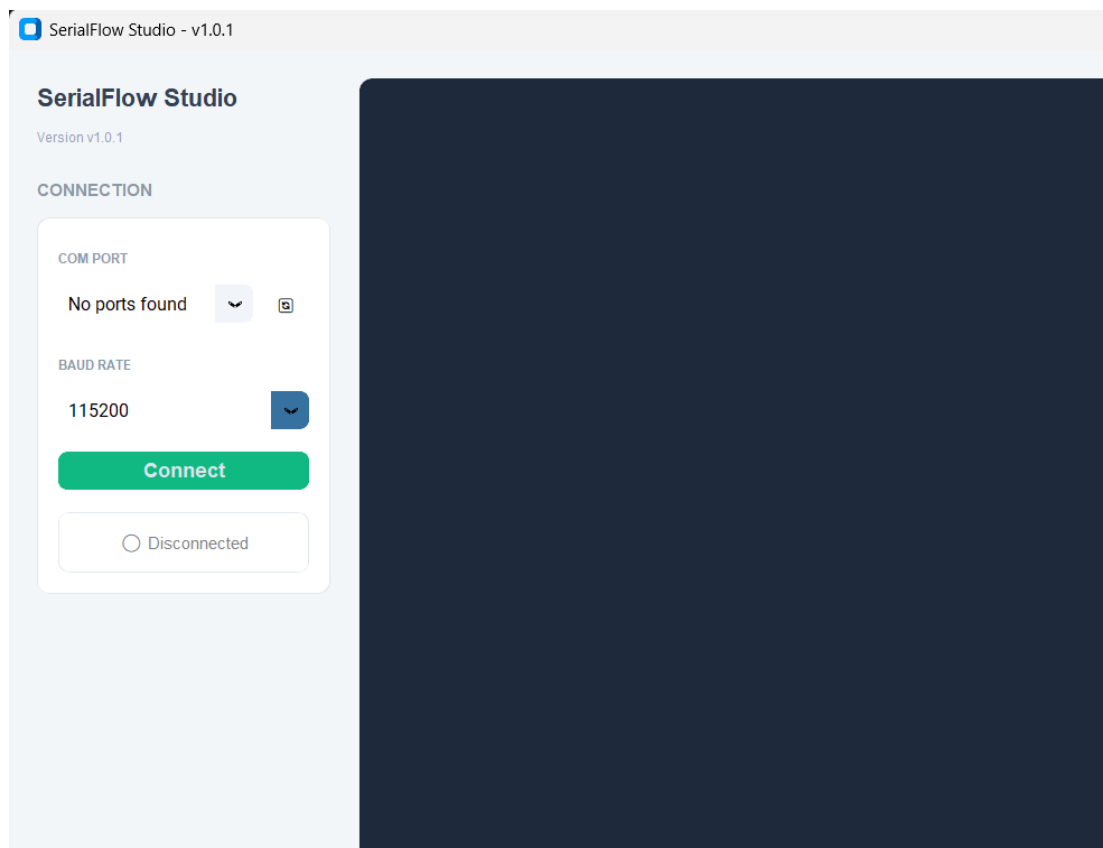


Figure 1. SerialFlow Studio main window with COM port selection, baud rate selection, connect control, status indicator, and terminal view.

1. Overview

SerialFlow Studio is a desktop tool for USB-to-serial communication. It is designed for firmware bring-up, debugging, command testing, and serial log capture on Windows systems.

Feature	What it does
COM port detection	Scans Windows serial ports and updates the dropdown when a USB serial adapter is plugged in or removed.
Baud rate selection	Supports common speeds such as 9600, 115200, and 921600 bps.
Terminal monitor	Displays received serial data in a dark terminal-style pane for easy reading.
Command send	Sends typed commands to the connected serial device with CR/LF line ending.
Timestamp option	Adds timestamp prefixes to received lines when enabled.
Log recording	Saves received terminal data to a user-selected text file.

2. Installation

Use these steps on a Windows machine with Python installed.

1. Open PowerShell in the SerialFlow_Studio project folder.
2. Install dependencies with: `pip install -r requirements.txt`
3. Connect the USB-to-serial adapter or target device to the PC.
4. Launch the tool with: `python usb_tool.py`

Driver note: If the COM port does not appear, install the correct Windows driver for the USB-to-serial chip used by the adapter or board.

3. Interface Walkthrough

- **COM PORT:** Select the detected serial port. Use the refresh control if you plug in a device after the tool is already open.
- **BAUD RATE:** Choose the serial speed expected by the target firmware or device.

- **Connect / Disconnect:** Opens or closes the serial connection. When connected, the selected port and speed are locked to avoid accidental changes.
- **Terminal view:** Shows incoming device data and sent commands.
- **Autoscroll:** Keeps the terminal pinned to the newest data.
- **Timestamp:** Adds a date and time prefix to new received lines.
- **Start Recording:** Prompts for a log file path and writes received data into the file until recording is stopped.

4. Connecting to a Device

1. Plug in the USB-to-serial adapter or target board.
2. Select the COM port from the COM PORT dropdown.
3. Select the baud rate that matches the device firmware.
4. Click Connect.
5. Confirm that the status changes to Connected and the terminal begins showing device output if the target is transmitting.

5. Sending Commands

After the connection is open, type the command in the input field and press Enter or click Send. SerialFlow Studio sends the command with a carriage return and newline so command-line style firmware shells can process it.

6. Recording Logs

1. Click Start Recording.
2. Choose the folder and log file name.
3. Keep the serial session running while data is collected.
4. Click Stop Recording to close the file safely.

Logging scope: The log file records received terminal data. Keep Autoscroll enabled during active debug sessions if you want the live view to follow new data.

7. Build the Windows EXE

The repository includes a PyInstaller spec file and this build command.

```
python -m PyInstaller --noconsole --onefile --collect-all customtkinter --name
"SerialFlow_Studio" usb_tool.py
```

After the build completes, the executable is created in the dist folder. Build and dist outputs are intentionally ignored by Git so the repository stays focused on source files and documentation.

8. Troubleshooting

Issue	Likely cause	Fix
No ports found	Adapter not connected, missing driver, or device disabled.	Reconnect the adapter, install the chip driver, then refresh the COM port list.
Cannot connect	Wrong port, port already open in another terminal, or unsupported baud rate.	Close other serial tools, confirm the COM number in Device Manager, then reconnect.
Unreadable characters	Baud rate mismatch or firmware uses a different serial format.	Match the baud rate to the firmware and confirm 8-N-1 serial settings on the target.
Connection lost	USB adapter unplugged or target reset.	Reconnect the device and click Connect again after the COM port appears.
Log file empty	Recording started after data was already received or connection was inactive.	Start recording before the test run and verify the connection is active.

9. Repository

Repository	https://github.com/satish2707/SerialFlow_Studio
Branch	main

Primary source file

usb_tool.py

10. Quick Test Checklist

COM port appears after connecting the USB-to-serial adapter.	Selected baud rate matches the target device.
Connect button opens the port without errors.	Incoming data appears in the terminal view.
Command send writes data to the device.	Recording creates a readable log file.